

A Finalized Algorithm for Curve Sketching

1. Determine the domain of the function. [Any value for which the function is undefined creates a discontinuity - hole or asymptote]
2. Determine the y-intercept, and, if easy to do, any x-intercept(s)
3. a) Determine any vertical asymptotes.
b) Use left- and right-sided limits to determine behaviour near asymptote
4. a) Determine any horizontal (use $\lim_{x \rightarrow \infty} f(x)$) or oblique (use long division) asymptotes
b) Consider the function as $x \rightarrow \infty$ and as $x \rightarrow -\infty$ to determine behaviour near horizontal or oblique asymptote.
5. a) Find $f'(x)$
b) Solve $f'(x)=0$ or $f'(x)$ DNE to find critical values
c) Create an interval chart to find intervals of increase and decrease.
d) Determine where $f(x)$ has local maximum and minimum points
6. a) Find $f''(x)$
b) Solve $f''(x)=0$ or $f''(x)$ DNE
c) Create an interval chart to find intervals of concavity.
d) Determine where $f(x)$ has any points of inflection
7. You may want to combine the two interval charts.
8. Sketch and fully label the graph of $f(x)$